LARGE LANGUAGE MODEL (LLM)

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INTRODUCTION TO AI, ML, AND LLMS

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Where do LLMs fit in this hierarchy…
  - LLMs are based mainly on **deep learning** and **neural networks**, especially a model architecture called the **Transformer**.

INTRODUCTION TO LLM

- **LLM** is a type of **Artificial Intelligence (AI)** model that is trained on huge amounts of text data so it can **understand, generate, summarize, translate, and reason using human language**.
- Examples:
  - ChatGPT
  - Gemini
  - Perplexity
  - Grok
  - Claude
  - Shazam

WHAT CAN LLMS DO?

- Text generation
- Code generation
- Summarization
- Question answering
- Translation
- Chatbots

LLMS WORKING MECHANISM

- Training on large text datasets
- Learning patterns, grammar, and meaning
- Predicting the **next word/token**

TRAINING PROCESS

- Pre-training vs Fine-tuning
- Supervised learning
- Reinforcement Learning from Human Feedback (RLHF)

TRANSFORMER ARCHITECTURE (CONCEPTUAL)

- Neural networks overview
- Transformer architecture
- Tokens, embeddings
- Attention mechanism
- Focuses on important words in a sentence

PROMPT ENGINEERING

- Definition
- Zero-shot, one-shot, few-shot prompting
- Examples of good vs bad prompts

TRADITIONAL NLP VS LLMS

Traditional NLP

- Rule-based
- Task-specific
- Limited context

LLMs

- Data-driven
- Multi-task
- Long context

ADVANTAGES & LIMITATIONS

Advantages

- Fast and scalable
- Can handle many task
- Improves productivity

Limitations & Risks

- Can give wrong answer
  -Not true ‘thinking’..
- Bias in data
- Privacy concerns

FUTURE OF LLMS

- Multimodal (text + image + audio)
- AI agents
- Smarter assistants
- Industry-wide impact

THANK YOU…

Any Queries?!…